

Development of reduced and optimized mechanism for ammonia/hydrogen mixture based on genetic algorithm

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Additional model validation

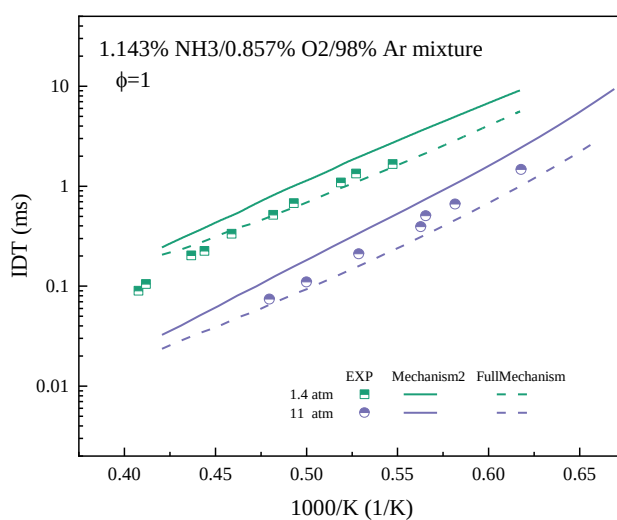


Figure S1 Measured (symbols) [1] and calculated (lines) IDTs of ammonia by optimized mechanism and initial mechanism under high temperature.

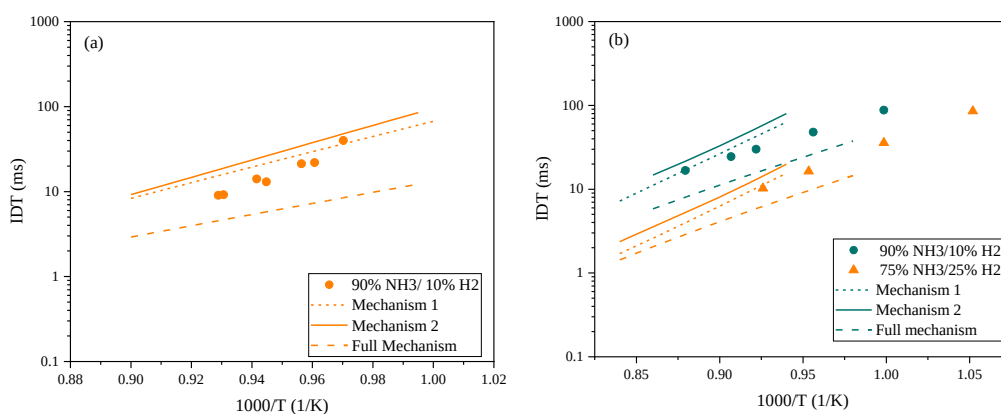


Figure S2 Measured (symbols) [2, 3] and calculated (lines) IDTs of ammonia/ammonia by reduced mechanism optimized mechanism and initial mechanism under low temperature and diluted condition, (a) $\phi=0.35$ and pressure of 43.4 bar and (b) $\phi=1$, pressure of 20 bar.

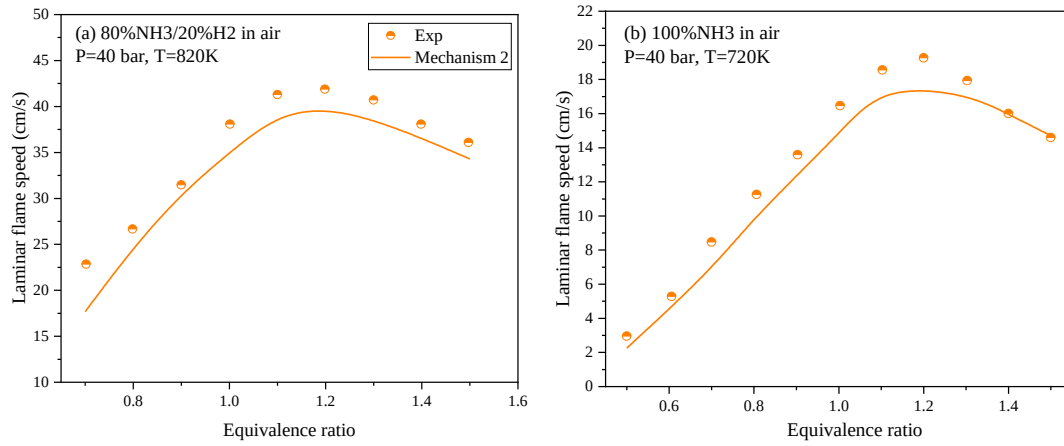


Figure S3 Measured [4] (symbols) and calculated (lines) LFSs by optimization mechanism and full mechanism under high temperature and high pressure.

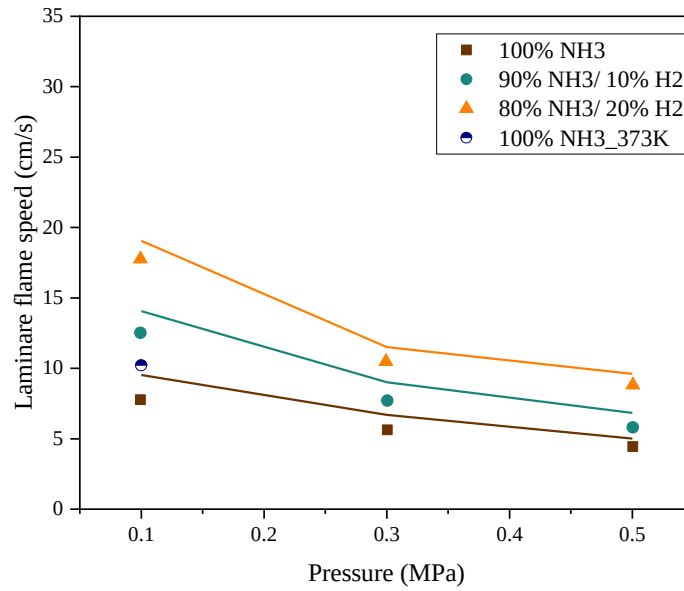


Figure S4 Measured [5, 6] (symbols) and calculated (lines) LFSs in air by optimization mechanism under temperature of 360K.

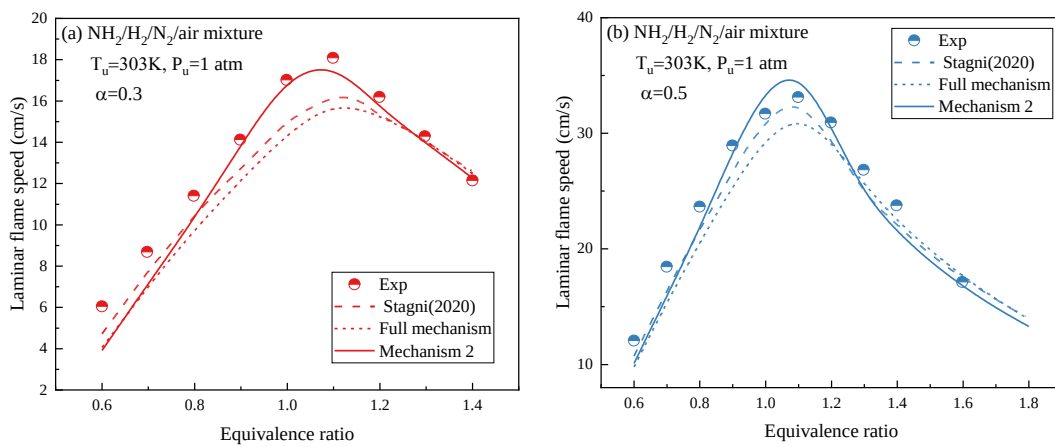


Figure S5 Comparison of Measured [7] (points) and calculated (lines) LFSs of NH₃/H₂/N₂/air mixture.

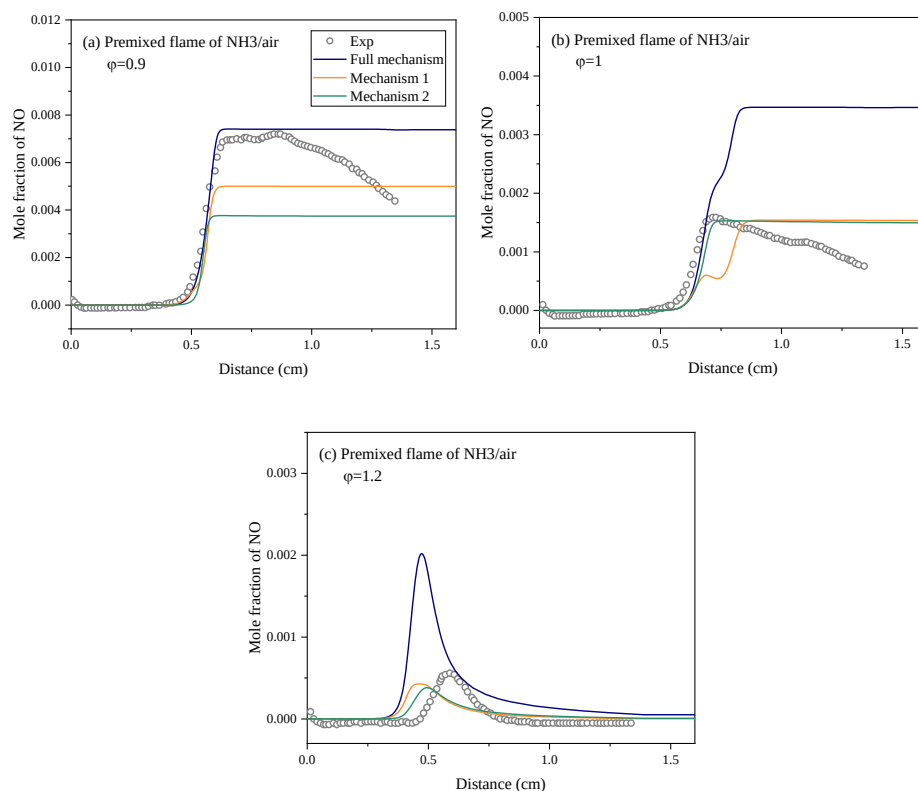


Figure S6 Mole fraction of NO in premixed flame of lean, stoichiometric, and rich NH₃/air mixture, experimental value [8].

Reference:

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